



Topic: Calculations with Matrices

Time: 45 mins

Marks: /45 marks

No calculator allowed

Question One: [1, 3: 4 marks]

If $A = \begin{bmatrix} 2 & -3 \\ 3 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 9 \\ 7 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 0 \\ 5 & 5 \end{bmatrix}$ evaluate:

a) $A + B$

b) $2A + 4B - C$

Question Two: [2, 2, 2, 2: 8 marks]

If $A = \begin{bmatrix} 4 & -2 \\ 6 & 1 \\ 7 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 4 \\ 5 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 1 & -1 & 4 \\ 9 & 10 & 3 \end{bmatrix}$ and $D = \begin{bmatrix} 1 \\ 4 \\ 6 \end{bmatrix}$ evaluate, where possible, each of the following. For those that cannot be calculated, clearly explain your reasoning.

a) AB

b) CA

c) B^2

d) DB

Question Three: [4, 2: 6 marks]

$$\text{If } A = \begin{bmatrix} 2 & 0 & 4 \\ 1 & 0 & 3 \\ -3 & 0 & -12 \end{bmatrix}$$

- a) Find $\mathbf{B}$ such that $\mathbf{A} + \mathbf{B} = \mathbf{I}$ (*The identity matrix*)
- b) Find $\mathbf{C}$ such that $\mathbf{A} - \mathbf{C} = \mathbf{O}$ (*The 3×3 zero matrix*) . Explain what you notice about matrix $\mathbf{C}$.

Question Four: [7 marks]

Find the values of the variables given that $\begin{bmatrix} m^2 - 2 & 3m + n \\ 4 & p^3 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 4 & 27 \end{bmatrix}$

Question Five: [1, 2, 4: 7 marks]

a) Find the value of P, a scalar, such that $P \begin{bmatrix} 4 \\ -3 \end{bmatrix} = \begin{bmatrix} -12 \\ 9 \end{bmatrix}$

b) Find the value of matrix Q such that $Q + \begin{bmatrix} 4 \\ -3 \end{bmatrix} = \begin{bmatrix} -12 \\ 9 \end{bmatrix}$

c) If $A = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$, show that $A^2 - 3A = \begin{bmatrix} -3 & 2 \\ -2 & -1 \end{bmatrix}$

Question Six: [2,2: 4 marks]

In the AFL preseason, points are awarded slightly differently. 6 points are awarded for a goal, 1 point for a behind and 9 points are awarded for a super goal, which is a goal scored from a distance greater than fifty metres.

At the end of the preseason;

Ricky had scored 12 goals, 30 behinds and 2 super goals,

Kevin had scored 9 goals, 18 behinds and no super goals and

Wayne had scored 10 goals, 9 behinds and 3 super goals.

- a) Construct an appropriate 3×3 matrix to multiply with $\begin{bmatrix} 6 \\ 1 \\ 9 \end{bmatrix}$ in order to calculate the total number of points scored by each player during the preseason.

- b) Hence calculate the total number of points scored by each player.

Question Seven: [3, 4, 2: 9 marks]

Jules runs a hot dog food van. Customers can order a hot dog on its own for \$3, a hot dog with a drink for \$4, or a hot dog, drink and chips for \$5. On a busy Saturday night he received 20 orders for hot dogs alone, 35 orders for hot dogs with a drink and 40 orders for a hot dog, drinks and chips.

- Show, using matrix calculations, Jules' total revenue for that particular Saturday night.
- If each hotdog costs Jules \$1.00, each drinks costs him \$0.50 and each serve of chips costs \$0.50, show using matrix calculations, the total costs of the products sold on that particular Saturday night.
- If Jules has no other expenses calculate his total profit for that Saturday night.